B-Series

HNW Vol. 1, sect. II.2-3

$$B(v, hf, u) = v(\emptyset)u + \sum_{\tau \in T} \frac{h^{|\tau|}}{\sigma(\tau)} v(\tau) F_f(\tau)(u)$$

The vectors F_f are associated with the ODE:

$$u'(t) = f(u) \quad (1)$$

$v(\tau)$: map from T to $\mathbb{R}$

h: step size

T: Set of all rooted trees

τ: a rooted tree

The series expresses the exact or numerical solution of (1) in terms of f and its derivatives, evaluated only at u_0.

$$B(v, hf, u_0) = v(\emptyset)u_0 + \sum_{\tau \in T} \frac{h^{|\tau|}}{\sigma(\tau)} v(\tau) F_f(\tau)(u_0)$$

In practice, we will work with two kinds of B-series.

B-series map: (discrete steps)

$$u_{n+1} = B(v, hf, u_n) \qquad v(\emptyset) = 1$$

B-series flow: (continuous flow)

$$u'(t) = B(v, hf, u) \qquad v(\emptyset) = 0$$

Derivatives of the true solution

$$u'(t) = f(u(t)) \qquad f : \mathbb{R}^m \to \mathbb{R}^m$$

$$u(t+h) = u(t) + hu'(t) + \frac{h^2}{2}u''(t) + \cdots$$
$$= \sum_{j=0}^{\infty} \frac{h^j}{j!} u^{(j)}(t)$$

1st derivative:

$$u_i'(t) = f^i(u(t))$$

2nd derivative:

$$u_i''(t) = \sum_j \frac{\partial f^i}{\partial u_j} u_j'(t) = \sum_j \frac{\partial f^i}{\partial u_j} f^j = f^i_j f^j$$

3rd:

$$u'''(t) = \sum_k \sum_j \frac{\partial f^i}{\partial u_j \partial u_k} f^j f^k + \sum_k \sum_j \frac{\partial f^i}{\partial u_j} \frac{\partial f^j}{\partial u_k} f^k$$

$$= f^i_{jk} f^j f^k + f^i_j f^j_k f^k$$

Here are all the terms we obtain up to 4th order, written in shorthand:

$$
\begin{tikzcd}[column sep=small]
 & & f^i \arrow[d] & & & \\
 & & f^i_j f^j \arrow[dl] \arrow[drr] & & & \\
 & f^i_{jk} f^j f^k \arrow[dl] \arrow[d] \arrow[dr] & & & f^i_j f^j_k f^k \arrow[dl] \arrow[d] \arrow[dr] & \\
f^i_{jk\ell} f^j f^k f^\ell & \underline{f^i_{jk} f^j_\ell f^k f^\ell} & \underline{f^i_{jk} f^j f^k_\ell f^\ell} & \underline{f^i_{j\ell} f^j_k f^k f^\ell} & f^i_j f^j_{k\ell} f^k f^\ell & f^i_j f^j_k f^k_\ell f^\ell
\end{tikzcd}
$$

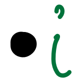
These are known as

Elementary Differentials

represented by

$F_f(\tau)$

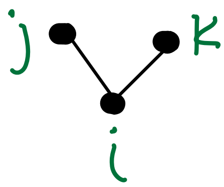
Rooted Trees

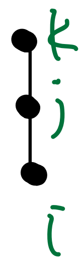


f^i

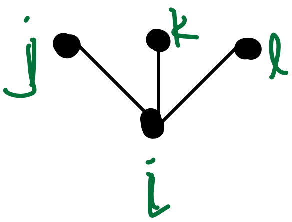
$f^i_j f^j$

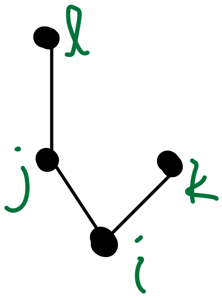
$f^i_{jk} f^j f^k$

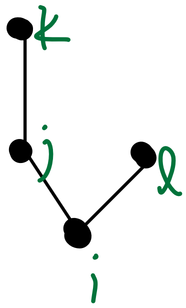


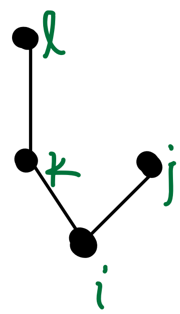


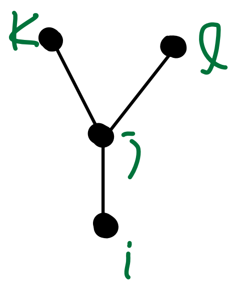
$f^i_j f^j_k f^k$

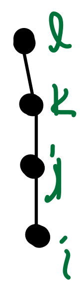






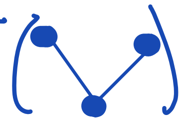






$f^i_{jkl} f^k f^l f^j$ $\quad$ $f^i_{jk} f^k f^j_l f^l$ $\quad$ $f^i_j f^j_{kl} f^k f^l$ $\quad$ $f^i_j f^j_k f^k_l f^l$

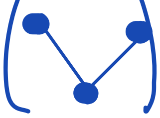
We write $F(\tau)$

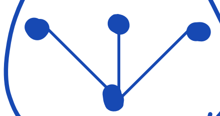
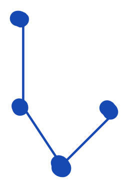
e.g. $F($  $) = f^i_{jk} f^j f^k$

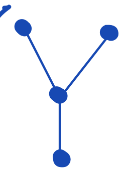
Butcher Series

$u(t_0) = F($$)(u^0)$

$u'(t_0) = F($$)(u^0)$

$u''(t_0) = F($$)(u^0) + F($$)(u^0)$

$u'''(t_0) = F($$)(u^0) + 3F($$)(u^0)$

$+ F($$)(u^0) + F($$)(u^0)$

We have

$$u^{(q)}(t_0) = \sum_{\tau \in LT_q} F(\tau)(u^0) = \sum_{\tau \in T_q} \alpha(\tau) F(\tau)(u^0)$$

labeled trees

unlabeled trees

Standard Taylor series:

$$u(t+h) = u(t) + hu'(t) + \frac{h^2}{2}u''(t) + \cdots$$
$$= \sum_{j=0}^{\infty} \frac{h^j}{j!} u^{(j)}(t)$$

So we get:

$$u(t+h) = \sum_{j=0}^{\infty} \frac{h^j}{j!} \sum_{\tau \in T_j} \alpha(\tau) F(\tau)(u(t))$$

Compare with our B-series:

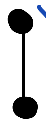
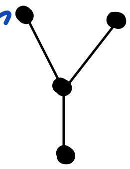
$$u(t+h) = u + \sum_{\tau \in T} \frac{h^{|\tau|}}{\sigma(\tau)} V(\tau) F_f(\tau)(u)$$

Equivalent if we take $\dfrac{V(\tau)|\tau|!}{\sigma(\tau)} = \alpha(\tau)$

Functions on trees

$\rho(\tau)$: order (# of nodes)

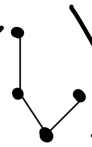
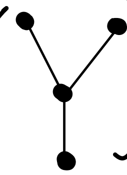
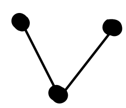
Also written as $|\tau|$

$\rho\Big($$\Big) = 2$ $\qquad$ $\rho\Big($$\Big) = 4$

$\gamma(\tau)$: density

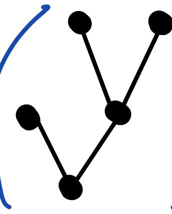
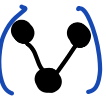
Also written as $\tau!$

$F(\tau)$: forest obtained by removing the root.

$F\Big($$\Big) = \Big\{$$,$$\Big\}$ $\qquad$ $F\Big($$\Big) =$ 

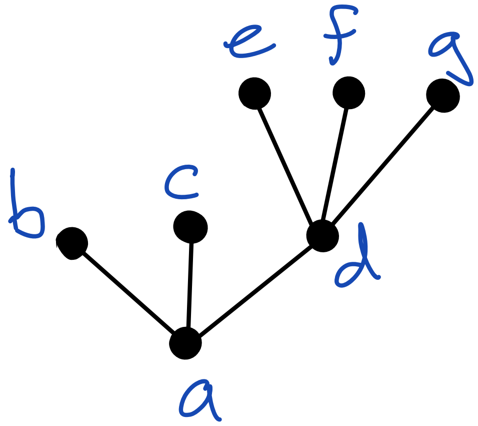
$$\gamma(\bullet) = 1$$

$$\gamma(\tau) = \rho(\tau) \cdot \prod_{\tau \in F(\tau)} \gamma(\tau)$$

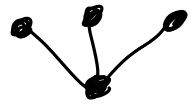
$\gamma($  $) = 5 \cdot \gamma(\bullet) \cdot \gamma($  $) = 5 \cdot 1 \cdot 3 \cdot 1 \cdot 1 = 15$

$\sigma(\tau)$: symmetry

The number of ways of permuting the nodes of the labelled tree τ such that the resulting tree without labels is identical.



$\sigma(\tau) = 2!\,3! = 12$



$\sigma(\tau) = 3!$

It turns out that

$$\alpha(\tau) = \frac{\rho(\tau)!}{\sigma(\tau)\gamma(\tau)}$$

Derivatives of the numerical solution

Now, we go back and compute derivatives of the Runge-Kutta solution

$$y_i = u^0 + h\sum_j a_{ij} f(y_j)$$

$$u' = u^0 + h\sum_j b_j f(y_j)$$

in the same way we did for the exact solution.

Note that $u' \approx u(h)$, so we compute derivatives of y_i w.r.t. h.

$$Y_i = U^n + h\sum_j a_{ij} f(Y_j)$$

Take Derivative of Jth Component of stage i

$$\left.\frac{dY_i^J}{dh}\right|_{h=0} = \sum_j a_{ij} f^J(Y_j) \cancel{+ h\sum_j a_{ij} \sum_K f_K^J(Y_j) \frac{dY_j^K}{dh}}$$

$$\left.\frac{d^2Y_i^J}{dh^2}\right|_{h=0} = 2\sum_j a_{ij} \sum_K f_K^J(Y_j) \frac{dY_j^K}{dh}$$

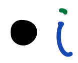
$$= 2\sum_j a_{ij} \sum_K f_K^J \sum_K a_{jK} f^K$$

$$\left.\frac{d(u^1)^J}{dh}\right|_{h=0} = \sum_j b_j f^J(Y_j)$$

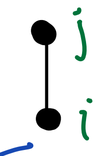
$$\frac{d^2(u^1)^J}{dh^2} =$$

Elementary Weights

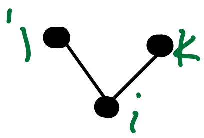
$c_i = \sum_j a_{ij}$



$\sum_i b_i$

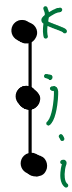


$\sum_{i,j} b_i a_{ij}$

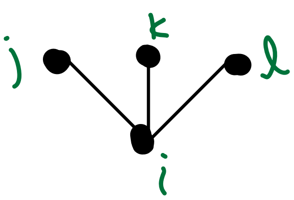


$\sum_{i,j,k} b_i a_{ij} a_{ik}$

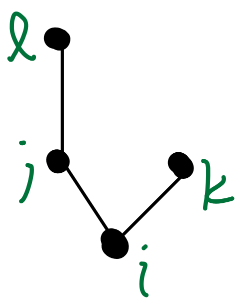
$= \sum_i b_i c_i^2$



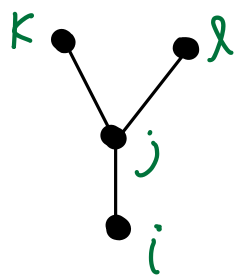
$\sum_{i,j,k} b_i a_{ij} a_{jk}$

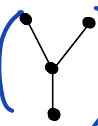


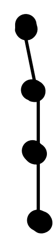
$\sum_{i,j,k,l} b_i a_{ij} a_{ik} a_{il}$

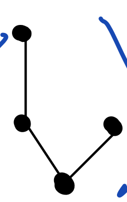


$\sum_{i,j,k,l} b_i a_{ij} a_{jl} a_{ik}$



$\sum_{i,j,k,l} b_i a_{ij} a_{jk} a_{jl} = \Phi\Big($  $\Big)$



$\Phi_i\Big($  $\Big) = \sum_{j,k,l} a_{ij} a_{jl} a_{ik}$

We get a B-series for the RK solution:

$$U^1(h) = \sum_{i=0}^{\infty} \frac{h^i}{i!} \sum_{\tau \in T_i} \alpha(\tau)\gamma(\tau)\sum_j b_j \underline{\Phi}_j(\tau) F(\tau)(U^0)$$

Order Conditions

Compare the B-series for the exact and numerical solutions:

$$u(t+h) = \sum_{j=0}^{\infty} \frac{h^j}{j!} \sum_{\tau \in T_j} \alpha(\tau) F(\tau) u(t)$$

$$u^{n+1} = \sum_{i=0}^{\infty} \frac{h^i}{i!} \sum_{t = T_i} \alpha(\tau) \gamma(\tau) \sum_j b_j \Phi_j(\tau) F(\tau)(u^n)$$

They match if we have

$$\sum_j b_j \Phi_j(\tau) = \frac{1}{\gamma(\tau)}$$

For a method of order p
we need

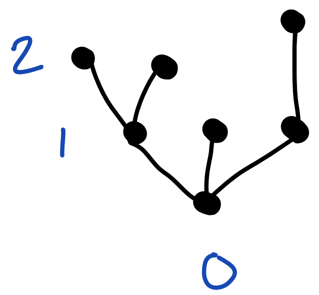
$$\sum_j b_j \Phi_j(\tau) = \frac{1}{\gamma(\tau)}$$

for all $\tau \in T_q \quad q = 1, 2, \ldots, p$

$$u'(t) = f(u)$$

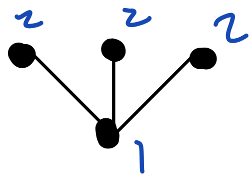
$$u'(t) = f(u) + hg(u)$$

Level sequence repr. of a rooted tree

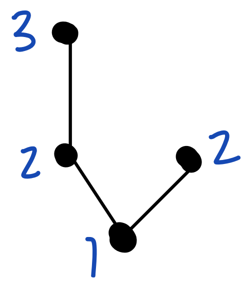


$[0, 1, 2, 2, 1, 1, 2]$

Level sequence

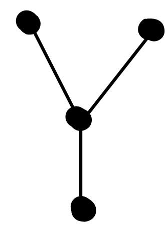


[1, 2, 2, 2]

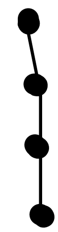


[1, 2, 3, 2]

~~[1, 2, 2, 3]~~

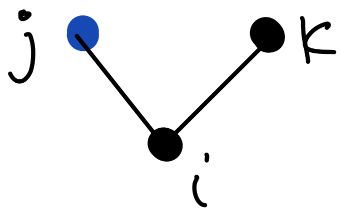


[1, 2, 3, 3]

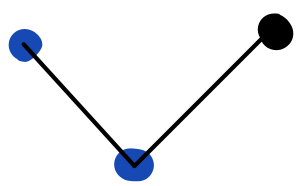


[1, 2, 3, 4]

$$u'(t) = f(u) + g(u)$$



$$f^i_{jk} g^j f^k$$



$$g^i_{jk} g^j f^k$$